Cole-Hopf Transformation for Compressible Navier-Stokes

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April 2026

Abstract

In the following paper we apply a Cole-Hopf transformation to the compressible Navier-Stokes equations to work towards a nonlinear vector Schrödinger equation. We make significant simplifications to the non-viscous isothermal compressible Navier-Stokes equations. We rewrite these equations as system of equations consisting of a density, a vector potential and a scalar potential evolution equation. We argue that the density equation cannot be turned into a Schrödinger type equation due to the lack of higher order derivatives of the density. Furthermore, we introduce two new potential Green's functions for the coupled nonlinearities in the vector potential and scalar potential evolution equations.

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1 Introduction

In the world of partial differential equations (PDEs) there are very equations that can be solved exactly. An approach that has worked for a variety of equations is to introduce a transformation that converts the PDE into one that can be solved more easily. An example of this is given by the Cole-Hopf transform that can transform the Burger's equation into the heat equation which gives us an exact solution, as shown in [1]. Moreover, it was shown by Okhitani and Vanon in [2] and [3] that the same transform could be applied to the incompressible Navier-Stokes equations to reduce them to a nonlinear Schrödinger equation.

Before jumping into any simplifications, it is important that we understand what it is we are simplifying. The Navier-Stokes equation gives us a system of equations for the evolution of a fluid in a region of space. It is to be noted that fluids in real life are made up of individual molecules, yet the system of equations are given as if the fluid were a continuum. We reason about this by noting that in a liter of water at standard conditions, there are around 55 moles of water molecules and hence $\sim 10^{25}$ particles. We therefore look to statistical mechanics to give us a way to deal with this, hence we look at aggregate behavior rather than simply individual behavior. A beautiful derivation done in [4] shows that taking statistical averages (moments) of the Boltzmann equation reduces it to the Navier-Stokes equations for single species nonrelativistic particles.

We will denote the density of the fluid by ρ and the velocity by $\mathbf{u}$. We note that these are functions of four variables, one time and three spatial, but we suppress the mention of these variables. Using the Helmholtz decomposition we note that $\mathbf{u}$, in the general setting, can be written as

$$\mathbf{u} = \mathbf{u}_c + \mathbf{u}_i$$

with $\nabla \cdot \mathbf{u}_c = 0$ and $\nabla \times \mathbf{u}_i = 0$. The first is called the incompressible mode and the second is called the compressible mode. Furthermore, we write the compressible mode in terms of a scalar function g for which $\mathbf{u}_c = \nabla g$ and the incompressible mode in terms of a vector potential such that $\mathbf{u}_i = \nabla \times \boldsymbol{\psi}$. The reason for this naming will be made more clear when we look at the incompressible Navier stokes equations.

With this notation established, we note that the zeroth moment of the Boltzmann equation leads to the conservation of mass equation

$$\partial_t \rho + \nabla \cdot (\rho \mathbf{u}) = 0 \tag{1}$$

The interpretation of this equation is that the change in density is given by the flow of density due to the velocity. The divergence of a vector valued function gives us the flux within a neighborhood of any point. Incompressibility is the assumption that $\nabla \cdot \mathbf{u} = 0$ which gives us that

$$\frac{D\rho}{dt} \equiv \partial_t \rho + \mathbf{u} \cdot \nabla \rho = 0$$

This is called the material derivative, and therefore incompressibility is also the assumption that the material derivative of ρ is 0.

Next we take a look at the first moment. This gives us the conservation of momentum equation, which we write as

$$\partial_t(\rho \mathbf{u}) + \nabla \cdot (\rho \mathbf{u} \otimes \mathbf{u}) + \nabla p = 2\nabla \cdot (\rho \nu \mathbb{S}) + \rho f$$

where p is the pressure of the fluid, ν is the viscosity of the fluid, $\mathbb{S}$ is the symmetric shear stress in the fluid given by

$$\mathbb{S}_{ij} = \frac{1}{2}(\partial_i u_j + \partial_j u_i) - \frac{1}{3}\delta_{ij}\partial_k u_k$$

and f is any external force acting on the fluid. For the remainder of the paper, we make the assumption that the fluid is isothermal. This means that $p = c_s^2 \rho$ where c_s is called the sound speed of the fluid. Moreover we assume that there are no external forces and no viscosity. This reduces our equations down to

$$\partial_t(\rho \mathbf{u}) + \nabla \cdot (\rho \mathbf{u} \otimes \mathbf{u}) + \nabla p = 0$$

which, as shown in [5][1], further reduces down to the form we will work with:

$$\begin{cases} \partial_t \boldsymbol{\psi} & = -\Delta^{-1}(\nabla \times (\mathbf{u} \times (\nabla \times \mathbf{u}))) \\ \partial_t(\mathbf{u}_i) + \partial_t(\mathbf{u}_c) & = -\partial_t(\Delta^{-1}(\nabla \times \nabla \times \mathbf{u})) - \Delta^{-1}\left(\partial_k \partial_j (u_i \partial_i u_j + \frac{1}{\rho}\partial_j p)\right) \end{cases} \tag{2}$$

Now equations (1) and (2) together form the Navier-Stokes equations for a compressible fluid with isothermal equation of state, no internal viscosity and no external forces acting on it:

$$\begin{cases} \partial_t \rho + \nabla \cdot (\rho \mathbf{u}) & = 0 \\ \partial_t \boldsymbol{\psi} & = -\Delta^{-1}(\nabla \times (\mathbf{u} \times (\nabla \times \mathbf{u}))) \\ \partial_t(\mathbf{u}_i) + \partial_t(\mathbf{u}_c) & = -\partial_t(\Delta^{-1}(\nabla \times \nabla \times \mathbf{u})) - \Delta^{-1}\left(\partial_k \partial_j (u_i \partial_i u_j + \frac{1}{\rho}\partial_j p)\right) \end{cases} \tag{3}$$

As is well known classical computation techniques for the Navier-Stokes equations are very computationally expensive and take a lot of time. We therefore look for transformations that convert them into a less computationally expensive set of equations. Recent work, as shown in [6], in solving the coupled nonlinear Schrödinger equations would suggest that a transformation converting the Navier-Stokes equations into this form would help, as effective schemes for numerical simulations for nonlinear Schrödinger equations are developed.

In what is to follow, we first give a description of the procedure as is applied to the compressible Navier-Stokes equations. We then use ideas from this procedure and apply it to the compressible Navier-Stokes equations under isothermal fluid conditions. In the compressible setting, we introduce a scalar potential as is guaranteed by the Helmholtz decomposition. This leads to significant simplifications in the Navier-Stokes equations. This allows us to see explicitly that in the limit where the compressible modes are very small, the equations reduce to the familiar incompressible Navier-Stokes equations. From there we apply the Cole-Hopf transformation. As we will see, the vector potential equation resembles a nonlinear vector Schrödinger equation, but more work is needed in order for the remaining equations to take suitable form.

[1] It is to be noted that this is not officially published work but rather the work of people I spoke with during this project.

2 Derivation of the Nonlinear Terms in the Schrödinger-Type Equations

Although we look specifically at the case of compressible fluid flow with no viscosity, it turns out that viscosity is important for reasons we will see below. To start, we look at the (incompressible) Navier-Stokes equations in terms of the vorticity, $\boldsymbol{\omega} = \nabla \times \mathbf{u}$, of the velocity field, $\mathbf{u}$. It is given by the following system of equations:

$$\begin{cases} \partial_t \boldsymbol{\omega} + (\mathbf{u} \cdot \nabla)\boldsymbol{\omega} & = (\boldsymbol{\omega} \cdot \nabla)\mathbf{u} + \nu \Delta \boldsymbol{\omega} \\ \nabla \cdot \mathbf{u} & = 0 \\ \mathbf{u}(\mathbf{x}, 0) & = \mathbf{u}_0(\mathbf{x}) \\ \boldsymbol{\omega}(\mathbf{x}, 0) & = \boldsymbol{\omega}_0(\mathbf{x}) \end{cases}$$

It is equivalent to the second equation above, if we had included the viscosity terms, as we will see.Using the incompressibility of the velocity field, we can find a divergence-free vector field $\boldsymbol{\psi}$ so that $\mathbf{u} = \nabla \times \boldsymbol{\psi}$. This comes from a choice of gauge, given by the uniqueness of solution to the Poisson equation. We call this the vector potential[2]. Now we find how the vorticity changes under this potential.

$$\begin{aligned} \boldsymbol{\omega} &= \nabla \times \mathbf{u} \\ &= \nabla \times (\nabla \times \boldsymbol{\psi}) \\ &= \nabla(\nabla \cdot \boldsymbol{\psi}) - \Delta \boldsymbol{\psi} \\ &= -\Delta \boldsymbol{\psi} \end{aligned}$$

Moreover consider the equation

$$\begin{aligned} \nabla \times (\mathbf{u} \times \boldsymbol{\omega}) &= \mathbf{u}(\nabla \cdot \boldsymbol{\omega}) - \boldsymbol{\omega}(\nabla \cdot \mathbf{u}) + (\boldsymbol{\omega} \cdot \nabla)\mathbf{u} - (\mathbf{u} \cdot \nabla)\boldsymbol{\omega} \\ &= (\boldsymbol{\omega} \cdot \nabla)\mathbf{u} - (\mathbf{u} \cdot \nabla)\boldsymbol{\omega} \end{aligned}$$

Using this allows us to rewrite the first equation in the Navier-Stokes equations above, in terms of $\boldsymbol{\psi}$, as

$$-\partial_t(\Delta \boldsymbol{\psi}) + (\mathbf{u} \cdot \nabla)\boldsymbol{\omega} - (\boldsymbol{\omega} \cdot \nabla)\mathbf{u} = \nu \Delta \Delta \boldsymbol{\psi}$$

Using what we derived above we get that this equation is of the form

$$-\Delta \boldsymbol{\psi}_t + \nabla \times (\mathbf{u} \times \boldsymbol{\omega}) = -\nu \Delta(\Delta \boldsymbol{\psi})$$

We can now get rid of one layer of the Laplacian by using Green's functions in the middle term. In doing so we get

$$-\boldsymbol{\psi}_t + g[\boldsymbol{\psi}] = -\nu \Delta \boldsymbol{\psi} \tag{4}$$

Where $g[\boldsymbol{\psi}]$ is a vector valued function that depends on $\boldsymbol{\psi}$ and so that $\Delta g_i[\boldsymbol{\psi}] = (\nabla \times (\mathbf{u} \times \boldsymbol{\omega}))_i$. We decipher this using index notation. Note that $\Delta g_i[\boldsymbol{\psi}] = \epsilon_{ijk}\partial_j(\mathbf{u} \times \boldsymbol{\omega})_k$. We now explore the $(\mathbf{u} \times \boldsymbol{\omega})_k$ term to get that

$$\begin{aligned} (\mathbf{u} \times \boldsymbol{\omega})_k &= \epsilon_{k\alpha\beta}\mathbf{u}_\alpha \boldsymbol{\omega}_\beta \\ &= \epsilon_{k\alpha\beta}\mathbf{u}_\alpha \epsilon_{\beta\delta\sigma}\partial_\delta \mathbf{u}_\sigma \\ &= (\delta_{k\delta}\delta_{\alpha\sigma} - \delta_{k\sigma}\delta_{\alpha\delta})\mathbf{u}_\alpha \partial_\delta \mathbf{u}_\sigma \\ &= \mathbf{u}_\alpha \partial_k \mathbf{u}_\alpha - \mathbf{u}_\alpha \partial_\alpha \mathbf{u}_k \\ &= \mathbf{u}_\alpha \partial_k \mathbf{u}_\alpha - \partial_\alpha(\mathbf{u}_\alpha \mathbf{u}_k) \end{aligned}$$

since we note that $\partial_\alpha \mathbf{u}_\alpha = 0$ by $\nabla \cdot \mathbf{u} = 0$. Plugging this back in we get that

$$\Delta g_i[\boldsymbol{\psi}] = \epsilon_{ijk}\partial_j(\mathbf{u}_\alpha \partial_k \mathbf{u}_\alpha - \partial_\alpha(\mathbf{u}_\alpha \mathbf{u}_k))$$

Let us now consider the first term, $\epsilon_{ijk}\partial_j(\mathbf{u}_\alpha \partial_k \mathbf{u}_\alpha)$. Note that if we swap $j \leftrightarrow k$, the derivative terms are symmetric under this exchange, however, ϵ_{ijk} picks up a sign, hence this term is 0. We then conclude that $\Delta g_i[\boldsymbol{\psi}] = -\epsilon_{ijk}\partial_j \partial_\alpha(\mathbf{u}_\alpha \mathbf{u}_k)$. Now to get a form of each g_i we use Green's functions to get[3]

$$g_i(\mathbf{x}) = \frac{1}{4\pi} \int_{\mathbb{R}^3} \epsilon_{ijk}\partial_j \partial_\alpha(\mathbf{u}_\alpha \mathbf{u}_k) \frac{1}{|\mathbf{x} - \mathbf{y}|} \, \mathrm{d}^3\mathbf{y}$$

[2] Although the choice is up to gauge we use the word *the*.

[3] We write $g_i(\mathbf{x})$ instead of $g_i[\boldsymbol{\psi}](\mathbf{x})$ to save space.

Furthermore, define $f_{abc} = \epsilon_{abk}\mathbf{u}_c\mathbf{u}_k$. Then the functions g_i are actually

$$g_i(\mathbf{x}) = \frac{1}{4\pi}\int_{\mathbb{R}^3} \partial_j\partial_\alpha f_{ij\alpha}\frac{1}{|\mathbf{x}-\mathbf{y}|}\,\mathrm{d}^3\mathbf{y}$$

This is an improper integral since there is a pole at $\mathbf{x} = \mathbf{y}$, so take a limit as follows

$$g_i(\mathbf{x}) = \lim_{\epsilon\to 0^+}\int_{\mathbb{R}^3\setminus B(\mathbf{x};\epsilon)} \partial_j\partial_\alpha f_{ij\alpha}\Gamma(\mathbf{x}-\mathbf{y})\,\mathrm{d}^3\mathbf{y}$$

where $\Gamma(\mathbf{x}) = \dfrac{1}{4\pi|\mathbf{x}|}$. Doing integration by parts we can move the partials on $f_{ij\alpha}$ to Γ. So

$$(\partial_j\partial_\alpha f_{ij\alpha})\Gamma = \partial_j(\partial_\alpha f_{ij\alpha}\Gamma) - \partial_\alpha(f_{ij\alpha}\partial_j\Gamma) + f_{ij\alpha}\partial_\alpha\partial_j\Gamma$$

Using the properties of Green's functions we get that, since the boundary terms above must vanish,

$$g_i(\mathbf{x}) = \lim_{\epsilon\to 0^+}\int_{\mathbb{R}^3\setminus B(\mathbf{x};\epsilon)} f_{ij\alpha}\partial_\alpha\partial_j\Gamma(\mathbf{x}-\mathbf{y})\,\mathrm{d}^3\mathbf{y}$$

now we split this integral up into cases when $j = \alpha$ and $j \neq \alpha$:

$$g_i(\mathbf{x}) = \lim_{\epsilon\to 0^+}\delta_{j\alpha}\int_{\mathbb{R}^3\setminus B(\mathbf{x};\epsilon)} f_{ij\alpha}\Delta\Gamma(\mathbf{x}-\mathbf{y})\,\mathrm{d}^3\mathbf{y} + \lim_{\epsilon\to 0^+}\int_{\mathbb{R}^3\setminus B(\mathbf{x};\epsilon)} f_{ij\alpha}\partial_\alpha\partial_j\Gamma(\mathbf{x}-\mathbf{y})\,\mathrm{d}^3\mathbf{y}$$

The first term has the Laplacian of Γ which is, by construction, $\delta(\mathbf{x}-\mathbf{y})$. Taking the derivatives of Γ gives us

$$g_i(x) = \delta_{j\alpha}f_{ij\alpha}(\mathbf{x}) + \lim_{\epsilon\to 0^+}\int_{\mathbb{R}^3\setminus B(\mathbf{x},\epsilon)} K_{j\alpha}f_{ij\alpha}\,\mathrm{d}^3\mathbf{y}$$

where

$$K_{j\alpha} = \frac{1}{4\pi}\frac{\delta_{j\alpha}|\mathbf{x}|^2 + 3x_jx_\alpha}{|\mathbf{x}|^5} \tag{5}$$

Moreover $\delta_{j\alpha}f_{ij\alpha} = 0$ since it is a product of a symmetric and an antisymmetric tensor. We therefore define the T operator as

$$T[\boldsymbol{\psi}] = \frac{3}{4\pi}\lim_{\epsilon\to 0^+}\int_{\mathbb{R}^3\setminus B(\mathbf{x},\epsilon)} \frac{(\mathbf{r}\times(\nabla\times\boldsymbol{\psi}(\mathbf{y})))(\mathbf{r}\cdot(\nabla\times\boldsymbol{\psi}(\mathbf{y})))}{|\mathbf{r}|^5}\,\mathrm{d}^3\mathbf{y} \tag{6}$$

This then makes (4) into the equation

$$\partial_t\boldsymbol{\psi} - T[\boldsymbol{\psi}] = \nu\Delta\boldsymbol{\psi} \tag{7}$$

All the simplifications done above allow us to do the Cole-Hopf transformation in a way that yields the desired result since we want to get a nonlinear vector Schrödinger equation out of this equation.

2.1 Cole-Hopf Transformation for the Incompressible Navier-Stokes Equations

We can now introduce the Cole-Hopf transform. This transformation is of the form $\boldsymbol{\psi}_i = k\ln\theta_i$ and [2] shows that (7) becomes

$$\begin{cases} \partial_t\theta_i & = \nu\Delta\theta_i + t_i\left[\dfrac{\nabla\theta_j}{\theta_j}\right]\theta_i \\ t_i\left[\dfrac{\nabla\theta_j}{\theta_j}\right] & = kT_i\left[\dfrac{\nabla\theta_j}{\theta_j}\right] - \nu\dfrac{|\nabla\theta_i|^2}{\theta_i^2} \end{cases} \tag{8}$$

where T_i is the ith component of T. For the case of compressible Navier-Stokes, the exact form of the vector potential equations is not important. What is important however, is that we understand where the T operator can be used to make simplifications. We can see above that the T operator is a Green's functions that resolves nonlinearities of the form $\nabla\times((\nabla\times\boldsymbol{\psi})\times\Delta\boldsymbol{\psi})$. We will make use of this fact below in the compressible case.

3 Compressible Navier-Stokes Equations and Simplifications

The compressible Navier Stokes equations are given by

$$\begin{cases} \partial_t\rho + \nabla\cdot(\rho\mathbf{u}) & = 0 \\ \partial_t(\rho\mathbf{u}) + \nabla\cdot(\rho\mathbf{u}\otimes\mathbf{u}) + \nabla p & = 2\nabla\cdot(\rho\nu\mathbb{S}) + \rho f \end{cases} \tag{9}$$

We recognize that the first of these equations as the conservation of mass of for the fluid and the second one as the conservation of momentum of the fluid. But as mentioned earlier, we will make use of these equations in the form of equation (3) since it is more explicit in terms of the potentials:

$$\begin{cases} \partial_t \rho & = \Delta^{-1}(\partial_j \partial_k((\partial_k \rho) u_j + \rho \partial_j u_k)) \\ \partial_t \boldsymbol{\psi} & = -\Delta^{-1}(\nabla \times (\mathbf{u} \times \boldsymbol{\omega})) \\ \partial_t(\mathbf{u}_i) + \partial_t(\mathbf{u}_c) & = -\partial_t(\Delta^{-1}(\nabla \times \nabla \times \mathbf{u})) - \Delta^{-1}\left(\partial_k \partial_j (u_i \partial_i u_j + \frac{1}{\rho}\partial_j p)\right) \end{cases} \tag{10}$$

Now since $\nabla \times \mathbf{u}_c = 0$ we have that there is a scalar function g such that $\mathbf{u}_c = \nabla g$. The second equation above can be used to simplify the third equations as

$$\partial_t \nabla g = -\nabla \times \partial_t \boldsymbol{\psi} - \partial_t(\Delta^{-1}(\nabla \times \nabla \times \mathbf{u})) - \Delta^{-1}\left(\partial_k \partial_j (u_i \partial_i u_j + \frac{1}{\rho}\partial_j p)\right) \tag{11}$$

Now let us look at $\nabla \times \nabla \times \mathbf{u}$. We get

$$\begin{aligned} \nabla \times \nabla \times \mathbf{u} &= \nabla(\nabla \cdot \mathbf{u}) - \Delta \mathbf{u} \\ &= \nabla(\nabla \cdot \mathbf{u}_i) - \Delta \mathbf{u}_c - \Delta \mathbf{u}_i \\ &= \nabla(\nabla \cdot \nabla g) - \Delta \mathbf{u}_c - \Delta \nabla g \\ &= \nabla \Delta g - \Delta \nabla g - \Delta \mathbf{u}_c \\ &= -\Delta \mathbf{u}_c \end{aligned} \tag{12}$$

and so $\Delta^{-1}(\nabla \times \nabla \times \mathbf{u}) = -\mathbf{u}_c$. Plugging this back in we find that

$$\begin{aligned} \partial_t \nabla g &= -\nabla \times \partial_t \boldsymbol{\psi} - \partial_t(\Delta^{-1}(\nabla \times \nabla \times \mathbf{u})) - \Delta^{-1}\left(\partial_k \partial_j (u_i \partial_i u_j + \frac{1}{\rho}\partial_j p)\right) \\ &= -\nabla \times \partial_t \boldsymbol{\psi} + \partial_t(\mathbf{u}_c) - \Delta^{-1}\left(\partial_k \partial_j (u_i \partial_i u_j + \frac{1}{\rho}\partial_j p)\right) \\ &= -\Delta^{-1}\left(\partial_k \partial_j (u_i \partial_i u_j + \frac{1}{\rho}\partial_j p)\right) \end{aligned} \tag{13}$$

This leads to the following system of equations

$$\begin{cases} \partial_t \rho & = \Delta^{-1}(\partial_j \partial_k((\partial_k \rho) u_j + \rho \partial_j u_k)) \\ \partial_t \boldsymbol{\psi} & = -\Delta^{-1}(\nabla \times (\mathbf{u} \times \boldsymbol{\omega})) \\ \partial_t \nabla g & = -\Delta^{-1}\left(\partial_k \partial_j (u_i \partial_i u_j + \frac{1}{\rho}\partial_j p)\right) \end{cases} \tag{14}$$

Our next task is to rewrite everything in terms of the potentials so we can do a Cole-Hopf transform on the potentials. Firstly, let us write all the equations in terms of the potentials, rather than in terms of $\mathbf{u}$. To do this, we note that

$$u_i = \epsilon_{ijk} \partial_j \psi_k + \partial_i g \tag{15}$$

Then since $\boldsymbol{\omega} = \nabla \times \mathbf{u}$ we get that $\boldsymbol{\omega} = -\Delta \boldsymbol{\psi}$. The system of equations now reads

$$\begin{cases} \partial_t \rho & = \Delta^{-1}(\partial_j \partial_k((\partial_k \rho) u_j + \rho \partial_j u_k)) \\ \partial_t \boldsymbol{\psi} & = \Delta^{-1}(\nabla \times ((\nabla \times \boldsymbol{\psi}) \times \Delta \boldsymbol{\psi})) + \Delta^{-1}(\nabla \times (\nabla g \times \Delta \boldsymbol{\psi})) \\ \partial_t \partial_k g & = -\Delta^{-1}(\partial_k \partial_j (\epsilon_{i\alpha\beta} \partial_\alpha \psi_\beta \partial_i u_j)) + \Delta^{-1}(\partial_k \partial_j (\partial_i g \partial_i u_j)) + \Delta^{-1}\left(\partial_k \partial_j \left(\frac{1}{\rho}\partial_j p\right)\right) \end{cases} \tag{16}$$

Before going further, we can make many simplifications to these equations. Let us start with the last equation.

3.1 Simplification of the Scalar Potential Equation

Note that taking the Laplacian of the scalar potential equation, we get that

$$\partial_t \Delta \partial_k g = -\partial_k \partial_j (\epsilon_{i\alpha\beta} \partial_\alpha \psi_\beta \partial_i u_j) + (\partial_k \partial_j (\partial_i g \partial_i u_j) + \partial_k \partial_j \left(\frac{1}{\rho}\partial_j p\right) \tag{17}$$

Taking another kth partial, or in other words the divergence of this equation, we get that

$$\partial_t \Delta \Delta g = -\Delta(\epsilon_{i\alpha\beta}(\partial_j \partial_\alpha \psi_\beta) \partial_i u_j) + \Delta(\epsilon_{i\alpha\beta} \partial_\alpha \psi_\beta \partial_i \Delta g) + \Delta(\partial_j (\partial_i g \partial_i u_j)) + \Delta \partial_j \left(\frac{1}{\rho}\partial_j p\right) \tag{18}$$

Now taking an inverse Laplacian we get that

$$\partial_t \Delta g = -\epsilon_{i\alpha\beta}(\partial_j\partial_\alpha\psi_\beta)\partial_i u_j + \epsilon_{i\alpha\beta}\partial_\alpha\psi_\beta\partial_i\Delta g + \partial_j(\partial_i g\partial_i u_j) + \partial_j\left(\frac{1}{\rho}\partial_j p\right) \tag{19}$$

Rewriting the components of the velocity in terms of the potentials as $u_j = \epsilon_{jcd}\partial_c\psi_d + \partial_j g$ gives

$$\partial_t \Delta g = -\epsilon_{i\alpha\beta}(\partial_j\partial_\alpha\psi_\beta)\partial_i(\epsilon_{jcd}\partial_c\psi_d + \partial_j g) + \epsilon_{i\alpha\beta}\partial_\alpha\psi_\beta\partial_i\Delta g + \partial_j(\partial_i g\partial_i(\epsilon_{jcd}\partial_c\psi_d + \partial_j g)) + \partial_j\left(\frac{1}{\rho}\partial_j p\right) \tag{20}$$

Focusing on the third term we note that

$$\partial_j(\partial_i g\partial_i(\epsilon_{jcd}\partial_c\psi_d + \partial_j g)) = \epsilon_{jcd}(\partial_j\partial_i g)\partial_i\partial_c\psi_d + \epsilon_{jcd}\partial_i g\partial_j\partial_i\partial_c\psi_d + (\partial_i\partial_j g)^2 + \nabla g\cdot\nabla\Delta g \tag{21}$$

Now we want to rewrite at least one of these terms in a more favorable manner and the most likely candidate for this is the last term. Note that

$$\Delta(fg) = f\Delta g + g\Delta f + 2\nabla f\cdot\nabla g$$

and hence

$$\nabla f\cdot\nabla g = \frac{1}{2}(\Delta(fg) - f\Delta g - g\Delta f)$$

this means that

$$\nabla g\cdot\nabla\Delta g = \frac{1}{2}(\Delta(g\Delta g) - (\Delta g)^2 - g\Delta\Delta g)$$

and so

$$\partial_j(\partial_i g\partial_i(\epsilon_{jcd}\partial_c\psi_d + \partial_j g)) = \epsilon_{jcd}(\partial_j\partial_i g)\partial_i\partial_c\psi_d + \epsilon_{jcd}\partial_i g\partial_j\partial_i\partial_c\psi_d + (\partial_i\partial_j g)^2 + \frac{1}{2}(\Delta(g\Delta g) - (\Delta g)^2 - g\Delta\Delta g) \tag{22}$$

which means that

$$\begin{aligned}\partial_t \Delta g = &-\epsilon_{i\alpha\beta}(\partial_j\partial_\alpha\psi_\beta)\partial_i(\epsilon_{jcd}\partial_c\psi_d) - \epsilon_{i\alpha\beta}(\partial_j\partial_\alpha\psi_\beta)(\partial_i\partial_j g) + \epsilon_{i\alpha\beta}\partial_\alpha\psi_\beta\partial_i\Delta g + \partial_j\left(\frac{1}{\rho}\partial_j p\right)\\ &+ \epsilon_{jcd}(\partial_j\partial_i g)(\partial_i\partial_c\psi_d) + \epsilon_{jcd}\partial_i g\partial_j\partial_i\partial_c\psi_d + (\partial_i\partial_j g)^2 + \frac{1}{2}\Delta(g\Delta g) - \frac{1}{2}(\Delta g)^2 - \frac{1}{2}g\Delta\Delta g\end{aligned} \tag{23}$$

The second term in the first line and first term in the second line are the same if we swap $i \leftrightarrow j$ and rename $\alpha \to c$ and $\beta \to d$. This reduces the equation slightly to

$$\begin{aligned}\partial_t \Delta g = &-\epsilon_{i\alpha\beta}(\partial_j\partial_\alpha\psi_\beta)\partial_i(\epsilon_{jcd}\partial_c\psi_d) + \epsilon_{i\alpha\beta}\partial_\alpha\psi_\beta\partial_i\Delta g + \partial_j\left(\frac{1}{\rho}\partial_j p\right)\\ &+ \epsilon_{jcd}\partial_i g\partial_j\partial_i\partial_c\psi_d + (\partial_i\partial_j g)^2 + \frac{1}{2}\Delta(g\Delta g) - \frac{1}{2}(\Delta g)^2 - \frac{1}{2}g\Delta\Delta g\end{aligned} \tag{24}$$

Also note that the first term in the second line is a product of a symmetric and antisymmetric term, hence it is 0. This can be seen noting that the epsilon term in antisymmetric in $j \leftrightarrow c$ and $\partial_i g\partial_j\partial_i\partial_c\psi_d$ is symmetric in those indices. So this further simplifies to

$$\begin{aligned}\partial_t \Delta g = &-\epsilon_{i\alpha\beta}\epsilon_{jcd}(\partial_j\partial_\alpha\psi_\beta)(\partial_i\partial_c\psi_d) + \epsilon_{i\alpha\beta}\partial_\alpha\psi_\beta\partial_i\Delta g + \partial_j\left(\frac{1}{\rho}\partial_j p\right)\\ &+ (\partial_i\partial_j g)^2 + \frac{1}{2}\Delta(g\Delta g) - \frac{1}{2}(\Delta g)^2 - \frac{1}{2}g\Delta\Delta g\end{aligned} \tag{25}$$

Now taking the inverse Laplacian we end with

$$\begin{aligned}\partial_t g = &\Delta^{-1}\left(-\epsilon_{i\alpha\beta}\epsilon_{jcd}(\partial_j\partial_\alpha\psi_\beta)(\partial_i\partial_c\psi_d) + \epsilon_{i\alpha\beta}\partial_\alpha\psi_\beta\partial_i\Delta g + \partial_j\left(\frac{1}{\rho}\partial_j p\right)\right)\\ &+ \Delta^{-1}((\partial_i\partial_j g)^2) + \frac{1}{2}g\Delta g - \frac{1}{2}\Delta^{-1}((\Delta g)^2) - \frac{1}{2}\Delta^{-1}(g\Delta\Delta g)\end{aligned} \tag{26}$$

Now let us pay attention to the first term of the above. Note that

$$\begin{aligned}\epsilon_{i\alpha\beta}\epsilon_{jcd}(\partial_j\partial_\alpha\psi_\beta)(\partial_i\partial_c\psi_d) &= \partial_j((\nabla\times\boldsymbol{\psi})_i)\partial_i((\nabla\times\boldsymbol{\psi})_j)\\ &= \mathrm{tr}\left((\nabla(\nabla\times\boldsymbol{\psi}))(\nabla(\nabla\times\boldsymbol{\psi}))^T\right)\\ &= (\nabla\cdot(\nabla\times\boldsymbol{\psi}))^2\\ &= 0\end{aligned}$$

This reduces the equations down to

$$\begin{cases} \partial_t \rho &= \Delta^{-1}(\partial_j \partial_k((\partial_k \rho) u_j + \rho \partial_j u_k)) \\ \partial_t \boldsymbol{\psi} &= \Delta^{-1}(\nabla \times ((\nabla \times \boldsymbol{\psi}) \times \Delta \boldsymbol{\psi})) + \Delta^{-1}(\nabla g \times \Delta \boldsymbol{\psi})) \\ \partial_t g &= \Delta^{-1}\left(\epsilon_{i\alpha\beta}\partial_\alpha \psi_\beta \partial_i \Delta g + \partial_j\left(\frac{1}{\rho}\partial_j p\right) - \frac{1}{2} g \Delta \Delta g - \frac{1}{2}(\Delta g)^2 + (\partial_i \partial_j g)^2\right) + \frac{1}{2} g \Delta g \end{cases} \tag{27}$$

We can also make the substitution that $p = c_s^2 \rho$ to get

$$\partial_j\left(\frac{1}{\rho}\partial_j p\right) = c_s^2 \partial_j\left(\frac{1}{\rho}\partial_j \rho\right) \tag{28}$$

$$= c_s^2 \Delta(\ln \rho) \tag{29}$$

and this makes the equation into

$$\begin{cases} \partial_t \rho &= \Delta^{-1}(\partial_j \partial_k((\partial_k \rho) u_j + \rho \partial_j u_k)) \\ \partial_t \boldsymbol{\psi} &= \Delta^{-1}(\nabla \times ((\nabla \times \boldsymbol{\psi}) \times \Delta \boldsymbol{\psi})) + \Delta^{-1}(\nabla g \times \Delta \boldsymbol{\psi})) \\ \partial_t g &= \Delta^{-1}\left((\nabla \times \boldsymbol{\psi}) \cdot \nabla \Delta g + c_s^2 \Delta(\ln \rho) - \frac{1}{2} g \Delta \Delta g - \frac{1}{2}(\Delta g)^2 + (\partial_i \partial_j g)^2\right) + \frac{1}{2} g \Delta g \end{cases} \tag{30}$$

3.2 Simplification of the Density Equation

Let us now look at the density equation. Taking the Laplacian we get

$$\partial_t \Delta \rho = \partial_j \partial_k((\partial_k \rho) u_j + \rho \partial_j u_k) \tag{31}$$

Expanding this we get

$$\begin{aligned} \partial_t \Delta \rho &= \partial_j \partial_k((\partial_k \rho) u_j) + \partial_j \partial_k(\rho \partial_j u_k) \\ &= \partial_j(u_j \Delta \rho + \partial_k \rho \partial_k u_j) + \partial_j(\partial_k \rho \partial_j u_k + \rho \partial_j \Delta g) \\ &= \Delta g \Delta \rho + u_j \partial_j \Delta \rho + \partial_j \partial_k \rho \partial_k u_j + \partial_k \rho \partial_k \Delta g + \partial_j \partial_k \rho \partial_j u_k + \partial_k \rho \Delta u_k + \partial_j \rho \partial_j \Delta g + \rho \Delta \Delta g \end{aligned} \tag{32}$$

The fourth term and the second last term are both $\nabla \rho \cdot \nabla \Delta g = \frac{1}{2}(\Delta(\rho \Delta g) - \rho \Delta \Delta g - \Delta g \Delta \rho)$ hence we get that

$$\partial_t \Delta \rho = \Delta(\rho \Delta g) + u_j \partial_j \Delta \rho + \partial_j \partial_k \rho \partial_k u_j + \partial_j \partial_k \rho \partial_j u_k + \partial_k \rho \Delta u_k \tag{33}$$

Now note that the third and fourth terms are the same under swapping the indices $j \leftrightarrow k$ and we expand this out as

$$\begin{aligned} 2\partial_j \partial_k \rho \partial_j u_k &= 2\nabla(\partial_k \rho) \cdot \nabla u_k \\ &= \Delta(u_k \partial_k \rho) - u_k \Delta \partial_k \rho - \partial_k \rho \Delta u_k \end{aligned} \tag{34}$$

But the last two terms in this expression are the second and last terms in the density equation, hence we get that

$$\partial_t \Delta \rho = \Delta(\rho \Delta g) + \Delta(u_k \partial_k \rho) \implies \partial_t \rho = \rho \Delta g + u_k \partial_k \rho \tag{35}$$

or in terms of the potentials

$$\partial_t \rho = \rho \Delta g + \epsilon_{k\alpha\beta} \partial_\alpha \psi_\beta \partial_k \rho + \partial_k g \partial_k \rho \tag{36}$$

Finally note that

$$\nabla \cdot (\rho \nabla g) = \rho \Delta g + \nabla \rho \cdot \nabla g$$

hence

$$\begin{cases} \partial_t \rho &= \nabla \cdot (\rho \nabla g) + (\nabla \times \psi) \cdot \nabla \rho \\ \partial_t \boldsymbol{\psi} &= \Delta^{-1}(\nabla \times ((\nabla \times \boldsymbol{\psi}) \times \Delta \boldsymbol{\psi})) + \Delta^{-1}(\nabla g \times \Delta \boldsymbol{\psi})) \\ \partial_t g &= \Delta^{-1}\left((\nabla \times \boldsymbol{\psi}) \cdot \nabla \Delta g + c_s^2 \Delta(\ln \rho) - \frac{1}{2} g \Delta \Delta g - \frac{1}{2}(\Delta g)^2 + (\partial_i \partial_j g)^2\right) + \frac{1}{2} g \Delta g \end{cases} \tag{37}$$

A quick glance at the density equation suggests that we cannot transform it into a Schrödinger type equation using a Cole-Hopf transform since we are missing second order derivatives in ρ. However, it is good to see that it separates itself very nicely into compressible and incompressible term. Moreover, looking at the vector potential equation we see that it separates into a coupled and non coupled term. It also has the property that the non coupled term is the exact same.

4 Cole Hopf Transform

Define the Cole-Hopf variables χ, θ, σ such that

$$\begin{aligned} \rho &= \ell \ln \chi \\ \psi_i &= n \ln \theta_i \\ g &= m \ln \sigma \end{aligned}$$

This makes the density equation transform as follows:

$$\frac{\ell}{\chi}\partial_t \chi = m\ell \frac{(1+\sigma)}{\sigma^2}\Delta\sigma \ln\chi + m\frac{\Delta\sigma \ln\chi}{\sigma} + \frac{m\ell}{\sigma\chi}\nabla\chi\cdot\nabla\sigma + n\ell\frac{1}{\theta_k\chi}\epsilon_{ijk}(\partial_i\chi)\partial_j\theta_k \tag{38}$$

As mentioned above, the density equation does not transform into a nonlinear Schrödinger equation since there is no Laplacian of the transformed variable. Now note that the first term in $\boldsymbol{\psi}$ equation transforms exactly the same as was mentioned in previous work, so we make use of the T operator we have defined earlier. However, the other terms are not like this. For this, we introduce a G operator, which is the Green's function for the coupled nonlinearity in the vector potential equation. But unlike the T operator we do not derive an explicit formula for this. So in this case, the $\boldsymbol{\psi}$ equation transforms as

$$\partial_t\theta = T\left[\frac{n\Delta\theta}{\theta}, \frac{n|\nabla\theta|^2}{\theta^2}\right]\theta + G\left[\frac{m\Delta\sigma}{\sigma}, \frac{n\Delta\theta}{\theta}, \frac{n|\nabla\theta|^2}{\theta^2}\right]\theta$$

We can suppress factors of n into the other operators. Unfortunately the scalar potential does not seem to have any curl of curl type equations and so we cannot make use of any previous logic with the T operator. For this reason we bundle this nonlinearity up into one operator F that we treat as a black box for now. In this case we get that

$$m\partial_t\sigma = F\left[\Delta\chi, \frac{\nabla\theta}{\theta}, \frac{\nabla\sigma}{\sigma}\right]\sigma + \frac{1}{2}m^2\ln\sigma\left(\frac{\Delta\sigma}{\sigma} - \frac{|\nabla\sigma|^2}{\sigma^2}\right)\sigma$$

The Cole-Hopf transformation now transforms the equations in (3) into the following system of equations

$$\begin{cases} \ell\partial_t\chi &= m\ell\chi\frac{(1+\sigma)}{\sigma^2}\Delta\sigma\ln\chi + m\chi\frac{\Delta\sigma\ln\chi}{\sigma} + \frac{m\ell}{\sigma}\nabla\chi\cdot\nabla\sigma + n\ell\frac{1}{\theta_k}\epsilon_{ijk}(\partial_i\chi)\partial_j\theta_k \\ \partial_t\theta &= T\left[\frac{n\Delta\theta}{\theta}, \frac{n|\nabla\theta|^2}{\theta^2}\right]\theta + G\left[\frac{m\Delta\sigma}{\sigma}, \frac{n\Delta\theta}{\theta}, \frac{n|\nabla\theta|^2}{\theta^2}\right]\theta \\ m\partial_t\sigma &= F\left[\Delta\chi, \frac{\nabla\theta}{\theta}, \frac{\nabla\sigma}{\sigma}\right]\sigma + \frac{1}{2}m^2\ln\sigma\left(\frac{\Delta\sigma}{\sigma} - \frac{|\nabla\sigma|^2}{\sigma^2}\right)\sigma \\ \chi(0) &= \exp\left(\frac{\rho(0)}{\ell}\right) \\ \theta(0) &= \exp\left(\frac{\boldsymbol{\psi}(0)}{n}\right) \\ \sigma(0) &= \exp\left(\frac{g(0)}{m}\right) \end{cases} \tag{39}$$

5 Discussion and Suggested Future Work

Our first order of business is to decipher whether or not G and F are even well defined operators. A similar approach to the derivation of the T operator will have to be taken since G amounts to finding the Green's function of the PDE given by $\Delta G = \nabla g \times \Delta\boldsymbol{\psi}$. Finding a form of F may require a different direction of simplification since the current approach was inspired by looking for explicit mentions of Δg. So another approach to this problem might be to look into the simplification of the scalar potential and not force there to be a Laplacian term. This might yield better results in terms of the simplifications for the F operator.

In terms of alternative approaches, for the density equation we might have more luck doing an "inverse" Cole-Hopf transformation in the sense that we want $\chi = \ell\ln\rho$ instead of $\rho = \ell\ln\chi$. This is motivated by the fact that this is the term that shows up in the scalar potential equation and therefore might lead to further simplifications.

Even after resolution of this operator we see that the vector potential equation is only *almost* a Schrödinger equation, but we are missing Laplacian terms. We suspect that this is due to the viscosity free condition since viscosity was needed in the incompressible case. This motivates a further step of looking into what adding in the viscosity term entails.

As mentioned earlier as well, it might be interesting to look into the weakly compressible limit. This is to ensure that we are on the right track and the manipulations we have done still reduce the cases we are familiar with.

Moreover we would also be interested in coupling the Navier-Stokes equations with the Maxwell equations in order to get the Magnetohydrodynamical equations. This will allow us to simulate a larger variety of fluids, including magnetized fluids which are common place in astrophysics applications.

Finally, we would want to make sure that these schemes actually translate into a speedup in computation times for these equations. We would therefore want to compare the derived equations in comparison to already established classical computation schemes.

6 Conclusion

Starting from the Navier-Stokes equations we rewrote them in terms of a vector and scalar potential. We then made a myriad of simplifications to get them in a form in which we could apply the Cole-Hopf transform of the respective dynamical variables. Furthermore, we showed that the density equation cannot be rewritten as a nonlinear vector Schrödinger equation due to the lack of higher order derivatives of density. The vector potential equation, up to the absence of Laplacian due to no viscosity, is a nonlinear vector Schrödinger equation. The scalar potential equation has nonlinearities that cannot be resolved in their current form and the inclusion of stresses and viscosity is required. Moreover we hinted at the introduction of two new Green's functions that aid in computation and simplification to get the Navier-Stokes equations into coupled nonlinear vector Schrödinger equations.

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